

NeuFlow v3: Replacing the Convex Upsampler with an Implicit Neural Decoder

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Abstract—Note: This is a preliminary proposal. The project is actively in progress and the ideas, experiments, and results are continuously evolving.

I built NeuFlow v3 by swapping the convex upsampler in NeuFlow v2 [1] with a small coordinate-conditioned MLP, borrowing the idea from InfiniDepth [2], which does the same thing for depth estimation. The decoder takes multi-scale features and a pixel coordinate as input and predicts a residual flow correction on top of what the GRU already outputs. I have trained it on VKITTI2 so far (2100 pairs) and it is not beating v2 yet. The model overfits quickly at this data scale. The architecture itself works fine; the bottleneck is training data.

Index Terms—optical flow, implicit neural representation, real-time inference, deep learning, coordinate networks

I. INTRODUCTION

NeuFlow v2 [1] runs a CNN encoder, cross-attention matching, and a GRU refinement loop, then recovers full-resolution flow using convex upsampling: blending nearby coarse values with learned weights. That works, but it cannot produce sharp motion boundaries because you are fundamentally limited to blending neighbors.

InfiniDepth [2] takes a different approach for depth: a small MLP is queried at any pixel coordinate, conditioned on latent features, and predicts the value there directly. Boundaries come out sharper. I am applying the same idea to optical flow in NeuFlow v3, with one addition: warped frame-1 features are also fed in so the decoder has cross-frame correspondence information, which a single-image depth model does not need.

II. WHAT CHANGED FROM V2

A. Architecture

The backbone (CNN encoder, cross-attention, GRU refinement) is completely frozen and unchanged. Only the upsampling head is replaced. This was not the original plan: I tried end-to-end training with differential learning rates (backbone 1×10^{-5} , decoder 2×10^{-4}) and it completely diverged, with EPE oscillating between 1 px and 100 px throughout training. InfiniDepth needed 800k steps on 8 GPUs to stabilize joint training. With a 30k step budget on one GPU it does not work. Frozen backbone is the only stable option at this compute scale.

B. Hierarchical Gated Feature Fusion

The decoder gets features from three points in the frozen backbone:

- $\mathbf{f}_{\text{ctx}} \in \mathbb{R}^{64}$: context features from frame 0 at $\frac{1}{8}$ res;

- $\mathbf{f}_{s8} \in \mathbb{R}^{128}$: matching features at $\frac{1}{8}$ res;
- $\mathbf{f}_{s16} \in \mathbb{R}^{128}$: matching features at $\frac{1}{16}$ res.

These are fused hierarchically (shallow to deep) using the gated residual scheme from InfiniDepth:

$$\mathbf{z} = \sigma(\mathbf{W}_g \mathbf{f}_{\text{deep}} + \mathbf{b}_g) \odot \mathbf{f}_{\text{shallow}} + \mathbf{f}_{\text{deep}}, \quad (1)$$

The fused feature, plus the normalized query coordinate (u, v) and warped frame-1 features, goes into a 3-layer MLP that outputs a residual correction $\Delta \mathbf{f}$. The output layer is zero-initialized, so at step 0 the model is identical to v2.

C. Training Setup

Loss is sparse ℓ_1 over 4096 randomly sampled coordinates per image, which avoids the memory cost of dense supervision. Decoder lr: 2×10^{-4} , batch size 4, 10k steps on VKITTI2 (2100 training pairs).

III. RESULTS SO FAR

v3 is not beating v2 yet. Table I shows the gap on VKITTI2 Scenes 18 and 20 (1174 validation pairs). At batch size 4 over 2100 pairs, 10k steps is already 19 epochs. The model peaks there and then overfits, as the checkpoint sweep in Table II shows. Training longer consistently makes it worse. FlyingChairs training is the obvious fix.

TABLE I
VKITTI2 EVALUATION (SCENES 18 & 20)

Model	Mean EPE	1px Acc.	3px Acc.
NeuFlow v2 [1]	2.23 px	46.2%	78.4%
NeuFlow v3 (10k steps)	3.15 px	7.2%	71.1%

TABLE II
NEUFLOW V3 CHECKPOINT SWEEP (MEAN EPE)

Step	Mean EPE
5k	3.29 px
10k	3.15 px
15k	3.47 px
20k	3.16 px
25k	3.56 px
30k	3.69 px

Scale ablation. I also tested a 2-scale decoder (without $\mathbf{f}_{s16}$): best EPE was 3.20 px, marginally worse than the 3-scale version (3.15 px). The difference is small enough that

the training data size is clearly the limiting factor, not the number of feature scales.

IV. PROPOSED NEXT STEPS

Training data. The standard curriculum for optical flow (FlyingChairs, then FlyingThings3D, then VKITTI2) should fix the overfitting problem. This is the most important next step, and I expect EPE to drop below the v2 baseline once the decoder has seen enough variety.

Benchmarks. Once properly trained, evaluation on MPI-Sintel and KITTI-15 would give a clearer picture of where v3 actually stands relative to the field.

MLP ablation. The 2-vs-3-scale comparison is already done (Section III). Still need to check whether MLP depth and hidden width matter once training data is no longer the bottleneck.

End-to-end fine-tuning. Jointly training the backbone and decoder is likely the highest-upside direction, but it requires multi-GPU compute. On one GPU with a 30k step budget it does not converge.

High-Performance Compute (HPC) access. This is the critical blocker. Running FlyingChairs/Things3D training and any serious end-to-end fine-tuning is beyond what a personal workstation can handle. I am asking for guidance on accessing the university cluster or any institutional compute allocation to unblock these next steps.

V. CONCLUSION

I replaced the convex upsampler in NeuFlow v2 with a coordinate-conditioned implicit decoder and got the training stable under a frozen backbone. The decoder runs and produces reasonable flow, but it is not beating v2 yet because 2100 training pairs is not enough. That is a data problem, not an architecture problem. The next step is clear: train on FlyingChairs and FlyingThings3D, which needs HPC access I do not have right now. This document captures where things stand. The project is still actively developing and I am looking for feedback on direction, compute resources, and whether this is worth pushing toward a publication.

REFERENCES

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